El Niño and the California Water Year: A Descriptive Analysis of ENSO Intensity and Statewide Precipitation

DATASCI 203 Final Project (Group 2)

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August 4th, 2026

Introduction

Rainfall in California varies from year to year with weather patterns in the tropical Pacific. The El Niño–Southern Oscillation (ENSO), a coupled pattern of tropical Pacific ocean temperatures and air pressure, is the one climate signal widely believed to foreshadow the state's wet and dry years.

This report asks the following research question.

> **What is the relationship between the intensity of ENSO and total statewide precipitation in California over the corresponding water year?**

Describing this relationship is valuable on its own terms. It tells the public and planners in government and utilities (the Bureau of Reclamation, the California Department of Water Resources, and PG&E) what history actually says about ENSO. Prior research documents the qualitative pattern, in which El Niño tends to bring wetter-than-normal conditions to California and La Niña the inverse (Cayan, Redmond & Dettinger 1999, Redmond & Koch 1991, Schonher & Nicholson 1989), with stronger associations in Southern California (Jong, Ting & Seager 2016). Yet the relationship is famously non-deterministic, and recent years were unexpectedly wet despite La Niña (Guirguis et al. 2024). Our contribution is to put statewide, water-year-level numbers on that pattern. We deliberately make no causal or predictive claims.

Data

Provenance

The analysis draws on three publicly available series, stored raw in `/src/` and merged into a single canonical analysis file by `src/data/make_analysis_data.R`. Variable-level documentation lives in the project data dictionary, `src/data_dictionary.csv` (sources and pipeline in the appendix). The unit of observation is the **water year** (WY), the hydrologist's accounting year running October 1 through September 30. WY 1950–2025 gives $n = 76$ observations.

- **California statewide precipitation** (`precip_in`) is total water-year precipitation in inches, from NOAA's National Centers for Environmental Information *Climate at a Glance* statewide series (12-month totals ending September), which spatially averages quality-controlled station observations across California. The raw download, comment header and all, is preserved at `src/data_rainfall.csv`.

- **Oceanic Niño Index** (`oni_djf`) is NOAA Climate Prediction Center's ONI, the three-month running mean of sea-surface-temperature anomalies in the Niño 3.4 region, retrieved with the `rsoi` R package (`rsoi::download_oni()`) and averaged from December to February, the season centered inside each water year. NOAA's own phase classification (El Niño $\geq +0.5$, La Niña ≤ -0.5, otherwise Neutral) accompanies the index, along with strength bins, 0/1 phase indicators, a one year lag, and a linear time trend (`src/oni_djf_wateryear.csv`).
- **Southern Oscillation Index** (`rsoi`) is the November–March pressure-based SOI, sign-reversed so that, like the ONI, positive values indicate El Niño conditions. We use it as an alternative operationalization, not in the main models.

Index calculations. Both indices have precise operational definitions, reproduced here from NOAA's documentation. The ONI is the three-month running mean of sea-surface-temperature anomalies (ERSSTv5) averaged over the Niño 3.4 region of the equatorial Pacific (5°N to 5°S, 120°W to 170°W), with anomalies computed against centered 30-year climatologies that NOAA updates every five years so that long-run ocean warming is not misread as a permanent El Niño. `oni_djf` is the December–February value of that series. The SOI is pressure-based. NCEI defines it from monthly mean sea-level pressure (SLP) at Tahiti and Darwin, standardized as departures from the 1981–2010 base period (per NOAA NCEI, https://www.ncei.noaa.gov/access/monitoring/enso/soi).

$$\text{SOI} = \frac{Z_{\text{Tahiti}} - Z_{\text{Darwin}}}{\text{MSD}}, \qquad Z_s = \frac{P_s - \bar{P}_s}{\sigma_s},$$

where, for each station s (Tahiti or Darwin), P_s is the actual monthly SLP, $\bar{P}_s$ the base-period mean SLP, and

$$\sigma_s = \sqrt{\frac{\sum \left(P_s - \bar{P}_s\right)^2}{N}}, \qquad \text{MSD} = \sqrt{\frac{\sum \left(Z_{\text{Tahiti}} - Z_{\text{Darwin}}\right)^2}{N}},$$

with N the number of months. Negative SOI values mean below-normal pressure at Tahiti relative to Darwin and coincide with El Niño episodes, while sustained positive values coincide with La Niña. Because that orientation is opposite to the ONI's, `rsoi` is the November–March mean SOI multiplied by -1, so that positive values indicate El Niño conditions for both indices in this report.

The statewide series is not a single gauge but a spatial average over NOAA's GHCN-Daily ground network, and the network is worth seeing. Parsing the GHCN-Daily station metadata shows that 2,917 California stations have reported daily precipitation (median record 11 years), and between 476 and 1,218 of them have records spanning any given water year in our sample.

Each water year appears exactly once, but we flag a design deviation openly. These are 76 sequential measurements of one geographic unit (a time series) rather than the cross section the assignment prescribes. Our instructor approved this design. We mitigate it with Model 3 and check the residuals in the *Regression Assumptions* section. With $n = 76$, below the "hundreds" the assignment recommends, we do **not** rely on large-sample results alone. Following the assignment's guidance for smaller data sets, we evaluate the more stringent classical linear model assumptions in the *Regression Assumptions* section. The sample is also far too small to split into exploration and confirmation sets, so we instead limit the number of specifications examined and report every one of them, in the spirit of Unit 13's warning that unreported exploratory tests inflate false-positive rates.

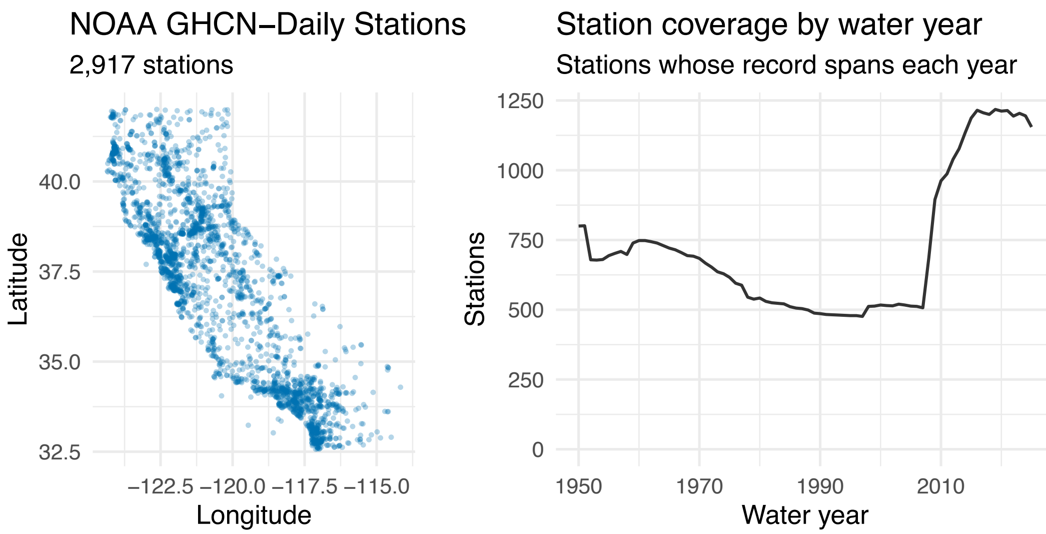

Figure 1: The measurement network behind the outcome variable. The left panel shows the California GHCN-Daily precipitation stations, which trace the state's shape and are densest along the populated coast and Central Valley. The right panel counts the stations whose precipitation record spans each water year.

Operationalization

Precipitation concept → `precip_in`. Statewide water-year totals are the standard measure for "how wet was California's year," and the water year cleanly contains one winter storm season.

ENSO intensity concept → `oni_djf`. We keep ENSO *metric* rather than collapsing it to categories, because intensity is the interesting part of the question ("strong El Niño" vs. "weak El Niño"). Between the two available indices we choose the ONI, because it is NOAA's operational definition of ENSO and is measured in physically interpretable units (°C of SST anomaly). The reversed SOI is a defensible alternative, and the two give nearly identical correlations with precipitation (0.257 vs. 0.260), but the ONI's phase thresholds are the ones used in public communication. The categorical phase labels are used only for display and group description, not as regressors. Table 1 compares the two measures. They never contradict each other outright (no year is El Niño under one and La Niña under the other), but they disagree on 27 of 76 years about whether a year passes the "Neutral" threshold, a reminder that phase *categories* are measurement choices, which is itself a reason to prefer the continuous index.

No observations were removed. All 76 water years join successfully and are complete on both analysis variables. The one structural exception is the previous-winter ONI used in Model 3, which is undefined for WY1950, the first year in the sample, so that specification fits on $n = 75$.

Table 1: Agreement between the two ENSO phase codings, WY 1950-2025. Rows show the reversed-SOI coding and columns the ONI coding. Off-diagonal years disagree only about the Neutral threshold.

	La Niña	Neutral	El Niño
La Niña	17	3	0
Neutral	12	13	8
El Niño	0	4	19

The two indices agree closely as metric measures (r = 0.87), so the disagreements counted in Table 1 reflect borderline winters near the category thresholds rather than conflicting signals.

The Shape of the Data

The rest of the report describes the joint distribution of winter ENSO intensity and water-year precipitation, starting from the raw ingredients. Table 2 shows the statistics for the outcome and the three ENSO measures.

Table 2: Descriptive statistics for the analysis variables, WY 1950-2025. The lagged ONI has 75 observations because it is undefined for the first sample year.

Variable	N	Mean	SD	Min	Median	Max
Precipitation (in)	76	22.18	6.75	10.86	20.80	40.41
ONI (Dec-Feb, °C)	76	0.01	1.03	-1.80	-0.15	2.60
Reversed SOI (Nov-Mar)	76	0.04	0.93	-2.15	0.14	3.16
Previous winter ONI (°C)	75	0.02	1.04	-1.80	-0.10	2.60

The full 76-year history complicates the expectation that El Niño years are wet years, because the wettest columns below are not all El Niño. The record winters of 1983 and 1998 were very strong El Niños, but 2017, the fourth-wettest year, was Neutral. The very strong El Niño winter of 2016 coincided with a roughly average year, and the driest year on record, 1977, was itself a weak El Niño.

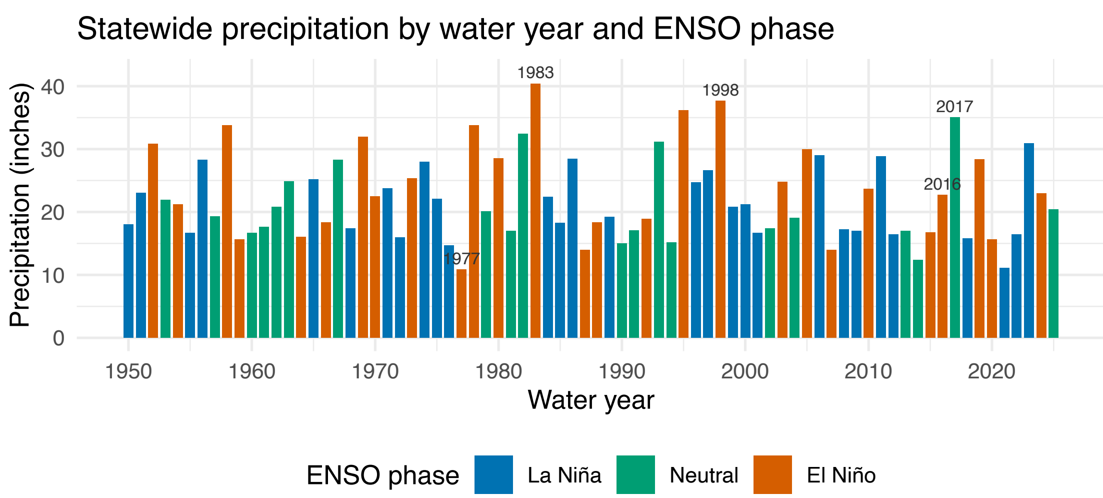

Figure 2: California statewide water-year precipitation, 1950-2025, colored by ENSO phase. The wettest years are not uniformly El Niño years, previewing the noisy relationship the models quantify.

Precipitation averages 22.2 inches with a standard deviation of 6.7. Year-to-year swings are enormous relative to the mean, which is exactly why a planning-relevant signal would matter. As the histograms above show, precipitation is right-skewed (skewness 0.65, falling to 0.11 after a log transform) while the winter ONI is roughly symmetric, so a transform is worth considering only on the outcome side. The skew is driven by a handful of very wet winters (1983, 1998, and 2017 among them). Table 3 summarizes precipitation by NOAA's phase labels.

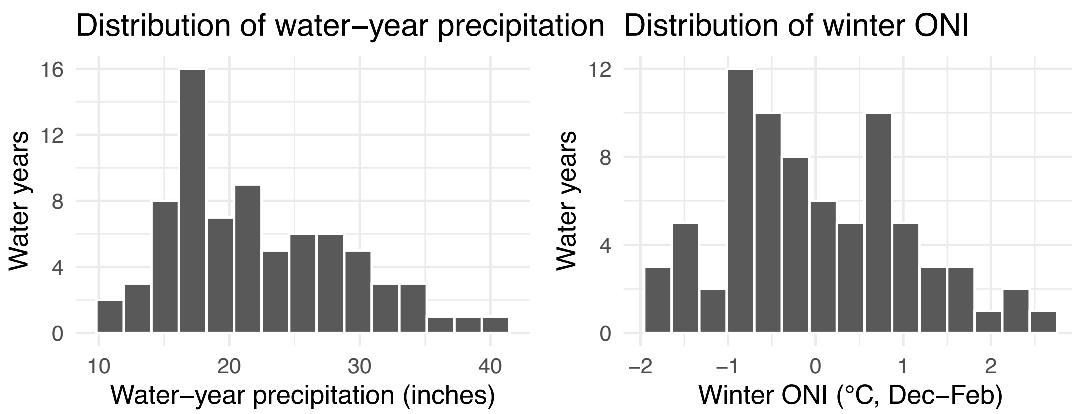

Figure 3: Water-year precipitation is moderately right-skewed (left), while the winter ONI is roughly symmetric with a long El Niño tail (right). The skew in precipitation motivates examining a log specification.

Table 3: Statewide water-year precipitation by ENSO phase. El Niño years average 3 or more inches wetter but are also markedly more variable.

ENSO phase (ONI, DJF)	Water years	Mean precip (in)	SD (in)	Driest (in)	Wettest (in)
La Niña	29	21.2	5.3	11.1	30.9
Neutral	20	20.9	6.3	12.4	35.0
El Niño	27	24.2	8.1	10.9	40.4

A simple comparison of group means offers only partial support for the expectation that El Niño years are wet years. On average, El Niño years received 3.1 more inches of precipitation than the other years taken as a group, but the difference is not statistically significant at the 5% level (Welch two-sample t-test, $|t| = 1.77$, two-sided $p = 0.084$). Because collapsing ENSO into categories discards information about event intensity, the regression models below work with the full metric index instead. Table 3 also shows that El Niño years are the most *variable* (SD 8.1 inches, versus roughly 5.8 for the other phases), a first sign of the heteroskedasticity the models will need to address.

The ENSO–Precipitation Relationship